System Analysis & Design: LinkedIn Job Market Analysis

Project: What Drives Demand and Pay for Data Roles | Track: Junior Data Analysis, DEPI | Version 1.0 | Phase 4 deliverable (milestone M2: ERD, data model, dashboard wireframes)

1. Project Overview

1.1 Problem

Job seekers, career advisors and recruiters cannot easily see in one place which roles are in demand, which skills and industries pay more, and how experience level, work type and competition change the picture. The raw data is a 123,849-posting LinkedIn snapshot (24 Mar to 20 Apr 2024) spread over 11 CSV files, with the postings file alone near 500 MB, so it cannot be used directly in Excel or read without a model.

1.2 Objectives

1. Clean and integrate the 11 source files into one star-schema model.
2. Measure demand by title, industry, company, location, work type and experience level.
3. Identify the most requested skills and benefits and relate them to salary.
4. Measure competition (views, applications, apply rate) per job category.
5. Deliver a 6-page Power BI dashboard with at least 7 documented, evidence-based insights, including a Data / Business Analyst deep-dive.

1.3 Scope

In scope: data cleaning, star-schema modelling, 25+ DAX measures, six dashboard pages, insights report, testing and GitHub documentation.

Out of scope: scraping or live data, predictive salary models, text mining of job descriptions, multi-year trends (the data covers about one month), Egypt-specific conclusions (the data is mostly US).

1.4 Stakeholders

Stakeholder	Interest	Main questions
Job seeker / fresh graduate	Which roles and skills to target	What do entry-level analyst jobs pay and require?
Career advisor / trainer	Evidence for teaching plans	Which skills are linked to higher pay?
Recruiter / HR analyst	Benchmarking	What salary, benefits and competition are typical for a role?
Project team and DEPI evaluators	Correct, reproducible work	Do totals reconcile with the source?

1.5 Expected Benefits

- One interactive view replaces manual exploration of 11 files.
- Skill and pay evidence helps learners choose what to study.
- Transparent coverage statements prevent misleading salary conclusions.
- A documented, reusable model and measure library demonstrates junior-analyst skills (cleaning, modelling, DAX, design).

1.6 Assumptions (labelled A1 to A8)

ID	Assumption
A1	Yearly salary is considered realistic between USD 20,000 and 500,000; the final range is confirmed after profiling and documented.
A2	Where currency is missing it is assumed USD; non-USD rows are excluded from salary analysis.
A3	Pay is annualised as hourly x 2,080, weekly x 52, biweekly x 26, monthly x 12.
A4	Posting date is derived from listed_time (milliseconds since 1970).
A5	Data Analyst roles are identified by title keywords: strict group (title contains Data Analyst, 408 jobs) and broad group (title contains Analyst, 4,920 jobs).
A6	Apply rate = applies / views, calculated only where applies is present and views > 0.
A7	The dashboard is read-only for viewers; there is no login or row-level security because the data is public.
A8	The team has three members, so some roles are combined (see section 4.3).

2. Requirements Analysis

2.1 Functional Requirements (summary of FR-1 to FR-12)

ID	Requirement	Priority
FR-1	Load and combine all 11 CSV files into one model	Must
FR-2	Clean data: remove duplicates, convert timestamps, standardise pay to yearly, flag salary outliers	Must
FR-3	Overview page with KPI cards (total jobs, companies, average and median salary, remote %)	Must
FR-4	Salary page by experience level, industry, state, work type, plus distribution	Must
FR-5	Skills & Benefits page: top skills, skills vs salary, common benefits	Must

ID	Requirement	Priority
FR-6	Industries & Companies page: top industries and employers, company size vs hiring	Must
FR-7	Demand & Competition page: views, applications, apply rate, sponsored vs non-sponsored	Must
FR-8	Data Analyst deep-dive page	Must
FR-9	Slicers for work type, experience level, industry, state, date	Must
FR-10	Drill-through from industry to company to job title	Should
FR-11	Tooltips showing salary coverage on salary visuals	Should
FR-12	Export of summary insights as a report	Could

2.2 Non-Functional Requirements

ID	Requirement	How the design meets it
NFR-1	Filters respond in under 5 seconds	Drop description text, keep needed columns, star schema, avoid heavy calculated columns
NFR-2	Totals reconcile with source (123,849)	DISTINCTCOUNT on job_id, reconciliation test
NFR-3	Usability	Consistent theme, KPI cards on top, navigation buttons
NFR-4	Reproducibility	Cleaning scripts and steps on GitHub
NFR-5	Transparency	Every salary visual shows coverage and filters
NFR-6	Privacy	Public, non-personal data only

2.3 Data Sources

All data comes from the public Kaggle dataset LinkedIn Job Postings (2023-2024).

Source table	Rows	Role in design	Key fields
postings	123,849	Fact	job_id, title, company_id, work_type, formatted_experience_level, remote_allowed, normalized_salary, views, applies, sponsored, listed_time, location
salaries	40,785	Pay detail (merged to fact)	job_id, min/med/max salary, pay_period, currency

Source table	Rows	Role in design	Key fields
job_skills	213,768	Bridge	job_id, skill_abr
job_industries	164,808	Bridge	job_id, industry_id
benefits	67,943	Fact-related	job_id, benefit type
companies	24,473	Dimension	company_id, name, size, state, country, city
company_industries	24,375	Company attribute	company_id, industry
company_specialities	169,387	Optional bridge	company_id, speciality
employee_counts	35,787	Company attribute	company_id, employee_count, follower_count
industries / skills	422 / 35	Lookups	id and name

2.4 Data Quality Constraints (from profiling)

Issue	Finding	Design response
Salary coverage	About 29% of postings	Analyse salary subset only; show coverage %
Salary outliers	Maximum in the hundreds of millions	Filter to documented yearly range (A1), keep an outlier flag
Remote flag	About 88% missing	Report as Not stated; no conclusions from this field alone
Applications	About 81% missing	Apply rate only where available
Timestamps	Milliseconds	Convert to date
Many-to-many links	Several skills and industries per job	Bridge tables and distinct counts
Geography	Mostly US	State the limit on the dashboard

2.5 Privacy and Security

- Only public data is used; no personal data (names of recruiters or applicants) is loaded.
- Company names are public business information and are retained.
- The Kaggle source is cited and its licence respected.
- The repository excludes the raw 500 MB file (use a download link and a small sample file instead) and contains no credentials.
- If published to Power BI Service, sharing is limited to project members, evaluators and a view-only link.

3. System Analysis

3.1 Current Process

Today, anyone wanting these answers must download 11 CSV files, open them separately (the large file often fails in Excel), match job_id and company_id by hand, and handle missing values and unit differences (hourly vs yearly pay) themselves. Results are not reproducible, many-to-many links are counted twice, and salary findings are quoted without noting that most postings have no salary.

3.2 User Needs

Persona	Need	Delivered by
Job seeker	Filter by experience level and see pay and top skills	Slicers, Salary page, Skills page, Data Analyst page
Career advisor	Compare skills by salary	Skills vs salary chart
Recruiter	Benchmark salary, benefits and competition by industry	Industries page, Benefits visual, Competition page
Evaluator	Trustworthy totals and documented method	Reconciliation checks, GitHub documentation

3.3 Proposed System: Inputs, Processes, Outputs

Stage	Description
Inputs	11 CSV files; lookup tables; cleaning rules (A1 to A6)
Processes	Profile, clean, standardise salary, convert dates, categorise titles, build star schema, write DAX measures, build visuals, test
Outputs	Cleaned tables, Power BI file (.pbix), 6-page dashboard, insights report (7+ insights), test report, presentation, GitHub repository

3.4 Feasibility

- **Technical:** Power BI Desktop and Python are free to use; 16 GB RAM is recommended, and the sample-then-full-refresh approach reduces the risk.
- **Schedule:** Milestones M1 to M5 run 16 Oct to 4 Dec 2026 with 6 to 8 hours per member per week.
- **Operational:** Read-only dashboard, no ongoing data feed, so maintenance is minimal.

4. System Design

4.1 Architecture

The system is a layered batch analytics pipeline:

```
[Source layer]      Kaggle CSVs (11 files)
        |
[Staging layer]     Python (pandas) cleaning scripts / Power Query
        |            -> cleaned CSV/Parquet files in /data/clean
[Model layer]       Power BI data model: star schema + bridges
        |            -> DAX measure library
[Presentation]      6-page Power BI dashboard (.pbix)
        |
[Delivery]          GitHub (docs, scripts, sample) + optional Power BI Service / PDF
export
```

4.2 Major Components

Component	Function
Cleaning scripts	Remove duplicates, annualise pay, flag outliers, convert dates, drop description text
Data model	Star schema with bridge tables for skills and industries
Measure library	Reusable DAX measures organised in a Measures table by folder
Dashboard	Six report pages, shared theme, slicer panel, navigation
Documentation	Data dictionary, cleaning log, insights report, test report

4.3 User Roles

Project roles (team):

Role	Responsibility
Project Manager	Plan, deadlines, GitHub submissions, review of each deliverable
Data Engineer	Cleaning scripts, outlier handling
Data Modeller and DAX Developer	Star schema, relationships, measures
Dashboard Designer and QA	Layout, theme, testing, presentation (shared between members per A8)

System roles (end users): Viewer (job seeker, advisor, recruiter) can filter, drill and export visuals; Author (team) can edit the model and report. No other roles are needed (A7).

4.4 Data Flow

1. Download CSVs and profile each table (rows, nulls, duplicates, ranges).
2. Python/Power Query cleans each table and writes clean files.
3. Power BI imports clean files; relationships are created.
4. DAX measures are written against the model.
5. Visuals consume the measures; slicers apply filter context.
6. Results are tested against Python/Excel checks, then published.

4.5 Recommended Technologies

Need	Tool	Reason
Cleaning large CSV	Python (pandas), Power Query	Handles 500 MB, repeatable scripts
Modelling and measures	Power BI Desktop, DAX	Taught in the track, strong star-schema support
Quick validation	Excel, pandas	Independent check of totals and medians
Version control	GitHub	Required for submission and reproducibility

4.6 Database (Data Model) Design

The model is a star schema. The fact table is one row per job; dimensions describe company, date, skill and industry.

fact_postings (grain: one row per job_id; target 123,849 rows)

Field	Type	Notes
job_id	Text/Int (PK)	Unique, no duplicates
company_id	Int (FK)	To dim_company; may be null
posted_date	Date (FK)	To dim_date, from listed_time
title	Text	Original title
title_group	Text	Derived: Data Analyst, Business Analyst, Other Analyst, Other
work_type	Text	Full-time, Part-time, Contract, etc.
experience_level	Text	Entry, Associate, Mid-Senior, etc.; Not stated if null
remote_allowed	Text	Remote, Not remote, Not stated
sponsored	Boolean	Sponsored posting flag
state, city, country	Text	Parsed from location

Field	Type	Notes
views, applies	Int	applies often null
salary_yearly	Decimal	Annualised, null if absent
salary_valid	Boolean	True if within range A1
has_salary	Boolean	Used for coverage %

dim_company: company_id (PK), company_name, company_size, state, country, city, employee_count, follower_count, primary_industry.

dim_date: date (PK), year, month, week, day, day_name (24 Mar to 20 Apr 2024).

dim_skill: skill_abr (PK), skill_name (35 rows).

dim_industry: industry_id (PK), industry_name (422 rows).

bridge_job_skill: job_id (FK), skill_abr (FK); one row per job-skill pair (213,768).

bridge_job_industry: job_id (FK), industry_id (FK) (164,808 rows).

fact_benefits: job_id (FK), benefit_type (67,943 rows).

bridge_company_speciality (optional): company_id (FK), speciality.

Relationships

From (one side)	To (many side)	Cardinality
dim_company.company_id	fact_postings.company_id	1 : many
dim_date.date	fact_postings.posted_date	1 : many
fact_postings.job_id	bridge_job_skill.job_id	1 : many
dim_skill.skill_abr	bridge_job_skill.skill_abr	1 : many
fact_postings.job_id	bridge_job_industry.job_id	1 : many
dim_industry.industry_id	bridge_job_industry.industry_id	1 : many
fact_postings.job_id	fact_benefits.job_id	1 : many

Skills and industries are many-to-many via the bridges. To let a skill or industry slicer filter the fact table, set cross-filter direction to Both on the bridge-to-fact relationship only (or use CROSSFILTER inside specific measures), and always count jobs with DISTINCTCOUNT(fact_postings[job_id]) so totals do not double count.

Text ERD

```
dim_date ---< fact_postings >--- dim_company
                  |
     +------------+-------------+
     |            |             |
bridge_job_skill  bridge_job_industry  fact_benefits
     |            |
  dim_skill   dim_industry
```

5. Data Analysis Workflow

5.1 Collection

Download the 11 CSV files from Kaggle, record the source and download date, and store them in /data/raw (excluded from GitHub; a 1,000-row sample is committed instead).

5.2 Cleaning Rules

Step	Rule	Example
Remove duplicates	Keep one row per job_id	Target: 0 duplicate job_id
Drop heavy columns	Remove description text	Cuts file size sharply
Convert timestamps	milliseconds to date	1713300000000 becomes a 2024 date
Annualise pay	Apply A3 multipliers	USD 40 per hour becomes 83,200 per year
Flag outliers	salary_valid = False outside A1 range	A salary of 200,000,000 is excluded from salary visuals but the job still counts for demand
Handle nulls	Replace missing categories with Not stated; leave numeric nulls	remote_allowed null becomes Not stated
Parse location	Split into city and state	Austin, TX becomes city Austin, state TX
Title grouping	Keyword rules (A5)	Title with Data Analyst becomes Data Analyst
Standardise text	Trim spaces, consistent case	Entry level and entry level are merged

Every step is logged (rows before and after) in a cleaning log kept on GitHub.

5.3 Transformation

- Merge salaries into the fact table by job_id (one row per job).
- Build dim_company by joining companies, company_industries and the latest employee_counts record per company.
- Create bridge tables from job_skills and job_industries, joined to their lookups.
- Load only required columns to protect memory (risk R1).

5.4 Core DAX Measures (examples from the 25+ library)

Measure	Logic
Total Jobs	`DISTINCTCOUNT(fact_postings[job_id])`
Total Companies	`DISTINCTCOUNT(fact_postings[company_id])`
Median Salary	`CALCULATE(MEDIAN(fact_postings[salary_yearly]), fact_postings[salary_valid] = TRUE())`
Average Salary	`CALCULATE(AVERAGE(fact_postings[salary_yearly]), fact_postings[salary_valid] = TRUE())`
Salary Coverage %	`DIVIDE(CALCULATE([Total Jobs], fact_postings[salary_valid] = TRUE()), [Total Jobs])`
Apply Rate	`DIVIDE(SUM(fact_postings[applies]), CALCULATE(SUM(fact_postings[views]), NOT(ISBLANK(fact_postings[applies]))))`
Remote %	`DIVIDE(CALCULATE([Total Jobs], fact_postings[remote_allowed] = Remote), [Total Jobs])` with the text literal quoted
% of Total Jobs	`DIVIDE([Total Jobs], CALCULATE([Total Jobs], ALL(fact_postings)))`
Skill Share	Jobs requiring the selected skill divided by Total Jobs
Data Analyst Jobs	`CALCULATE([Total Jobs], fact_postings[title_group] = Data Analyst)` with the text literal quoted

Additional measures: jobs per company, average views per job, sponsored share, benefit coverage, salary premium vs overall median, and similar.

5.5 Analytical Methods

Question	Method	Visual
Where is demand highest?	Counts and percent of total by title, industry, state	Bar chart, filled map
What drives pay?	Median salary grouped by experience, industry, work type, skill	Clustered bar, box-style column
Is pay spread wide?	Distribution with binned salary	Histogram (binned column)
Which skills pay more?	Median salary per skill with job counts	Bar with count labels, scatter (skill share vs median pay)
How competitive are jobs?	Views, applications, apply rate by category	Bar, scatter
Does company size matter?	Jobs and salary by size band	Column chart
Entry-level analyst picture	Compare Data Analyst group vs all roles	KPI cards, ranked bars

Medians are used rather than averages because salary is skewed; all salary results are reported with coverage (A1, NFR-5). Where groups have fewer than about 30 salary records, the visual is flagged as low sample size.

5.6 Presentation

Findings are presented in the dashboard and summarised in at least 7 documented insights, each with the visual used as evidence, the figure, and a limitation note. Example insight format: Mid-Senior roles in industry X show a median salary of Y on Z% coverage (salary subset only).

6. Interface Design

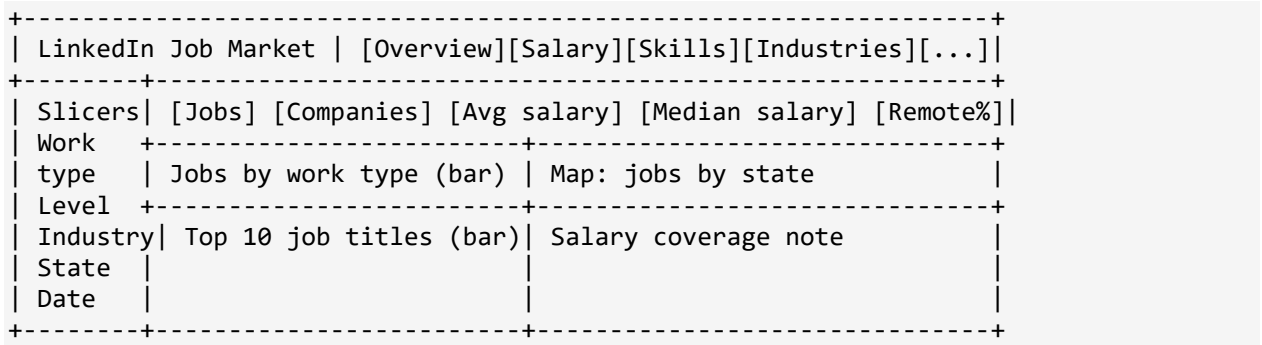
6.1 Global Layout

- Top bar: page title, coverage note, and navigation buttons to the six pages.
- Left or top slicer panel: work type, experience level, industry, state, date (FR-9), kept in the same position on every page.
- KPI cards on the first row of each page; one message per page (Few, Knaflic).
- Consistent palette (one accent colour for highlights, neutral grey for context), readable fonts and clear titles that state the message.

6.2 Pages

Page	Purpose	Main visuals
1. Overview	Market snapshot	KPI cards (jobs, companies, average and median salary, remote %), jobs by work type, top 10 titles, map of jobs by state
2. Salary	What drives pay	Median salary by experience level, industry, state, work type; salary histogram; coverage card
3. Skills & Benefits	In-demand skills	Top skills bar, skills vs median salary scatter, benefits bar
4. Industries & Companies	Who hires	Top industries, top employers, company size vs hiring, drill-through to company then title
5. Demand & Competition	Competition	Views, applications, apply rate by category, sponsored vs non-sponsored comparison
6. Data Analyst Deep-dive	Focus on target role	Data Analyst job count, salary by experience, top skills, top industries and employers, location split

6.3 Wireframe: Page 1 (Overview)



6.4 Interaction

- **Slicers** filter all visuals on a page and sync across pages where relevant.
- **Cross-highlighting:** clicking a bar highlights related values in other visuals.
- **Drill-through:** right-click an industry to open a company and job-title detail page (FR-10).
- **Tooltips:** hover on salary visuals shows job count and salary coverage % (FR-11).
- **Reset button** clears all slicers; **bookmarks** switch between default and Data Analyst-focused views.

7. Testing and Implementation

7.1 Testing Strategy

Level	What is tested	Example test and expected result
Data tests	Cleaning output	No duplicate job_id; all dates fall in 24 Mar to 20 Apr 2024; no salary_valid row outside range
Model tests	Relationships	Skill counts do not exceed total jobs when counted with DISTINCTCOUNT
Measure tests	DAX correctness	Median salary equals pandas median on the same subset
Reconciliation	Totals	Dashboard Total Jobs = 123,849
Visual tests	Layout and slicers	Each slicer filters every relevant visual on its page
Performance	Speed	Filter response under 5 seconds on full data
Usability	Navigation	Two peers complete three tasks (for example, find median entry-level salary) without help

Target: at least 95% of test cases passed, with failures logged, fixed and retested.

7.2 Implementation and Deployment

1. Develop on a sample (for example, 10,000 postings), then refresh with the full data.
2. Commit scripts, documentation, data dictionary, test report and screenshots to GitHub; the .pbix is added if file size allows (otherwise a download link).
3. Optionally publish to Power BI Service and share a view-only link, plus export the dashboard to PDF as a fallback.
4. Present the final dashboard on 4 Dec 2026.

7.3 Maintenance

- Keep the data dictionary and cleaning log updated when rules change.
- If a newer dataset version is used, re-run the scripts and repeat the reconciliation tests.
- Tag the final repository version (for example, v1.0) for the submission.

7.4 Risks (summary of the project risk plan)

Risk	Mitigation
Large file slows Power BI	Load needed columns only, sample then full refresh
Low salary coverage and outliers	Salary subset only, show coverage, filter by documented range

Risk	Mitigation
Missing remote and applies values	Document missing rates, avoid unsupported conclusions
One-month, US-centric snapshot	State limitations, avoid trend or Egypt-specific claims
Double counting from many-to-many links	Bridge tables, DISTINCTCOUNT, reconciliation test
Missed deadlines or unavailable member	Weekly check-ins, buffers, shared repository

7.5 Possible Future Improvements

- Mine job descriptions (NLP) for skills beyond the 35 standard categories.
- Add multi-period data to study trends over time.
- Build a simple salary-prediction model, with clear uncertainty.
- Add benchmarks for other countries, including Egypt, with suitable data.
- Automate refresh with a scheduled pipeline and add row-level security if private data is introduced.

8. Traceability Summary

Requirement	Design element
FR-1, FR-2	Sections 4.6, 5.2, 5.3
FR-3 to FR-8	Section 6.2 (pages 1 to 6)
FR-9 to FR-11	Sections 6.1 and 6.4
NFR-1, NFR-2	Sections 5.2, 5.4, 7.1
NFR-4, NFR-5, NFR-6	Sections 2.5, 5.2, 7.2
BR-1 to BR-5	Sections 5.5 and 6.2